

Local Rigidity Thesis — Extended ToC

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Chapter 1 — Introduction

Objective. Open with the framing observation, narrow to Kripke's case, and set up the dissertation's specific contribution.

1.1 The Problem: What Kripke Left Open

Kripke broke with descriptivism by treating proper names as rigid designators across all possible worlds. The semantic thesis is well-motivated and largely correct. What the *Naming and Necessity* lectures and the 1971 "Identity and Necessity" paper leave open is the cognitive story: what operation a speaker performs when rigidly tracking an individual across counterfactual variation, what baptism is at the level of cognition, and how causal chains of communication are psychologically realized. The SEP entry on rigid designators describes Kripke's appeal to stipulation as "more like a promissory note than the satisfaction of an explanatory obligation."

1.2 Thesis Statement

Rigid designation is the social stabilization of x^o -mode tracking across absence. The necessity of identity is a consequence of x^o -mode designation, not a property of identity itself. The necessary *a posteriori* arises because the same identity statement is necessary at the x^o level and discoverable at the x^c level. PRU supplies the cognitive mechanism Kripke's apparatus stipulates without explaining.

1.3 Methodology and Scope

The argument proceeds by close reading of Kripke's identity formula and its natural-language gloss, then through Russell and Wittgenstein, then through the constructive PRU proposal. The scope is metasemantic: the dissertation does not propose changes to Kripke's model theory; it offers a cognitive account of the preconditions Kripke's semantics presupposes.

1.4 Roadmap

Chapter 2 reads Kripke’s necessity-of-identity formula and exposes a three-way equivocation on “object.” Chapter 3 compresses the Russell–Kripke debate and locates the explanatory gap. Chapter 4 introduces the x^o/x^c distinction and shows what analytical work it does. Chapter 5 develops the PRU account proper: baptism as lifting, causal chains as testimonial networks, PRU as metasemantics. Chapter 6 addresses six objections. Chapter 7 reconstructs the argument and lists open questions.

Chapter 2 — Identity, Objects, and Designators

Objective. Starting from Kripke’s own necessity-of-identity formula, run through Wittgenstein, expose the extensional/intensional and variable/instance problems, and arrive at the diagnosis that the formula presupposes the designator-mode distinction.

2.1 Kripke’s Sentence

The natural-language gloss reads: “for every object x and object y , if x and y are the same, then it is necessary that x and y are the same object.” The word “object” is doing work. It transforms x and y from what they are in the formula — bound variables ranging over a domain under an assignment function — into something that sounds like a particular: a thing in the world, something one might encounter, point to, name.

Rewriting the gloss with a concrete term — “for every object dog_1 and object dog_2 ” — exposes the move: someone named two encounters, then discovered the identity. The formula has no resources to represent this discovery; the formula’s variables are assigned by a function and rigid by construction. Kripke’s insertion of “object” before each variable creates the bridge between formal triviality and philosophical substance.

2.1.1 The Benjamin Franklin passage

A cleaner exhibit from Kripke’s own text. He picks out two individuals by descriptions (“invented bifocals,” “first Postmaster General”), then concludes: “ x and y are both Benjamin Franklin, and it can certainly be necessary that Benjamin Franklin is identical with himself.” The mode-switch is on the page: description-identified instances are replaced by a name, and necessity is read off the name’s rigidity. In PRU notation, $\mathcal{R}(man_x^c, bifocals^c)$ vs. $\mathcal{R}(man_x^o, bifocals^c)$ — the shift from x^c to x^o makes the necessity trivial but renders the discovery invisible.

2.2 Wittgenstein: Identity Between Objects Is Either Nonsense or Empty

Tractatus 5.5303: “Roughly speaking, to say of two things that they are identical is nonsense, and to say of one thing that it is identical with itself is to say noth-

ing at all.” Russell called this “a destructive criticism from which there seems no escape.” Wittgenstein’s radical solution: eliminate the identity sign entirely. Identity of reference is expressed by using the same sign, not by a sign for identity.

The core insight is that identity is a relation between designators, not between objects. Russell makes the same observation: identity is needed “as between a name and a description or between two descriptions.” The Wittgenstein–Kripke connection on identity (as distinct from the much-discussed rule-following connection) is essentially unexplored in the secondary literature.

2.3 The Formula: Extensional Validity, Intensional Silence

How $\forall x \forall y (x = y \rightarrow \Box(x = y))$ is actually evaluated: exhaustive enumeration of extensional assignments. Variables under assignment are pure extensional pointers, rigid by construction. The formula is trivially valid at the extensional level. It has no resources to address the intensional dimension where Hesperus/Phosphorus lives.

Relation to Schwarz (2006): the present observation is upstream of his circularity diagnosis. Schwarz argues that Kripke’s defense of necessity of identity for rigid terms is circular; the present chapter argues something prior — that the formula’s validity does not extend to names without an unargued metasemantic premise.

2.4 Variables, Instances, Names: The Three-Way Equivocation

Three things are conflated under “object”:

- **Variables** — mathematical, rigid by assignment. They carry no encounter-history, no observational profile, no epistemic thickness.
- **Instances** — encountered particulars, rigid within the encounter. Carry perceptual content but only for the duration of the encounter.
- **Names** — socially stabilized, rigid across absence. The category Kripke’s argument is ultimately about.

The gloss bridges all three under “object.” The formula is valid for the first; the philosophical claim is about the third; the transition passes through the second without being made explicit.

2.5 What the Formula Presupposes

The formula is valid because variables are rigid by construction and domain elements are self-identical by set-theoretic axiom. When Kripke applies it to names, the transition requires the substantive thesis that names behave like variables — that they hold their referent fixed across possible worlds. The formula does not establish this. Kripke argues for it independently through modal intuition tests.

The natural-language gloss — with its pivotal word “object” — lends the thesis an air of logical inevitability that belongs only to the extensional level. The formula presupposes what a metasemantic theory is supposed to explain.

Garson (2024) shows that the quantified modal logics that validate the necessity of identity for rigid terms are “very different from ones Kripke developed,” with “serious costs whichever solution we attempt.” The technical non-triviality matches the metasemantic gap.

Chapter 3 — From Russell to Kripke

Objective. The semantic picture, compressed. Russell’s positive role; descriptivism’s modal failure; Kripke’s solution; the gap PRU fills.

3.1 Russell and Frege: Names as Disguised Descriptions

The descriptivist thesis: a proper name abbreviates a cluster of definite descriptions that uniquely picks out an individual. Russell’s deeper move within this tradition: genuine names require acquaintance; only demonstratives qualify. Russell discovers local rigidity — the referential certainty of demonstrative use under acquaintance — then retreats by treating it as a limiting case restricted to immediate sense data.

The retreat is forced by the Cartesian demand for infallibility: identity must be “exempt from all imaginable doubt.” Outside immediate sense data, doubt is possible; therefore (for Russell) rigid reference is impossible. The dissertation argues, against Russell, that rigidity does not require infallibility.

3.2 The Modal Failure of Descriptions

Descriptions pick out satisfiers world-by-world — constitutively world-relative. No accumulation of descriptive conditions produces cross-world tracking. The scope problem makes this formally explicit: $\iota x Fx$ relative to $\Box$ has primary and secondary occurrences with different truth conditions. Pulling $\iota x Fx$ outside the modal operator (large scope) mimics rigidity but only positionally — the description is evaluated once in the actual world, its denotation fixed, then the modal operator cannot re-run the sweep.

Kaplan’s *dthat* operator makes this explicit: $dthat(\iota x Fx)$ takes a description and rigidifies it. Large-scope $\iota x Fx$ and $dthat(\iota x Fx)$ do the same logical work. *Dthat* is nameification: converting derived rigidity into intrinsic rigidity by stipulation. The community recognized the gap within its own formal framework. Descriptions approximate rigidity only by becoming name-like; rigidity is the prior phenomenon.

3.3 Kripke’s Rigid Designation

Core definition: a designator is rigid if it picks out the same individual in every possible world in which that individual exists. *De jure* rigidity (the term is stipulated to refer to one individual) versus *de facto* rigidity (the description happens to pick out the same individual in every world).

The anti-descriptivist argument operates on three fronts:

- **Modal argument.** “Aristotle might not have taught Alexander” is intelligible. If “Aristotle” abbreviated “the teacher of Alexander,” the sentence would be a contradiction.
- **Epistemic argument.** Speakers can use a name competently without knowing any uniquely identifying description of the bearer.
- **Semantic argument.** The descriptions associated with a name can be largely false without the name failing to refer.

Reference is fixed by baptism and preserved through causal chains of communication: historical contact, not descriptive content.

3.4 The Explanatory Gap

What the semantic account leaves open: no cognitive story of rigid tracking, no cognitive mechanism for baptism, no psychological realization of the causal chain. Russell gestured at the answer (acquaintance) but couldn’t generalize beyond immediate sense data. Kaplan’s *dthat* formalizes nameification but stipulates rigidity by fiat.

The post-Kripkean literature acknowledges the gap openly. LaPorte’s “promissory note” formulation captures it: Kripke’s apparatus tells us *that* names are rigid, not *how* speakers achieve rigid tracking.

3.5 Existing Cognitive Implementations and Their Limits

Three serious attempts to fill the gap, each with a shared limitation:

- **Evans: discriminating conception.** The right desideratum — a speaker must be able to discriminate the bearer from other things — misdescribed as a “conception.” The word pulls the explanation back into the epistemic-propositional register.
- **Recanati: mental files.** Container metaphor — a file holds information about a bearer. Propositional, static.
- **Devitt: causal-descriptivism.** Closer than the others — keeps the causal chain as the reference-fixing mechanism — but still representationalist about what makes the chain work.

The shared limitation: all retain the template architecture in which a stored content is matched against perception. None explains the cognitive primitive from which rigid tracking emerges. The grouping of these three under one diagnosis is offered as the dissertation’s interpretive claim, not as consensus.

Chapter 4 — The x^o/x^c Distinction

Objective. Introduce the PRU framework through the distinction that does the work. This chapter carries the central analytical weight.

4.1 Two Modes of One Word

x^o marks a word coordinating with an experienced instance — “this x here.” x^c marks a word operating in linguistic space — “the x ,” “ x s are...” Same phonological token, two functional registers. Developmental evidence (Bloom, Tomasello): early word use is heavily anchored to encountered instances and only gradually generalizes into description-mode. The token “dog” pointing at a dog is dog^o ; the same token in “dogs are mammals” is dog^c .

This is not two homonymous words. It is one word in two modes. The mode is determined by use, not by lexical entry.

4.2 The Distinction Applied to Identity

At x^o : identity is either nonsense (two objects — Wittgenstein’s first horn) or trivial (one object — Wittgenstein’s second horn). Wittgenstein’s dilemma is correct for this level. There is nothing for identity to do: you are pointing at one thing, or you are pointing at two different things.

At x^c : identity is substantive and contingent. Two intensional profiles may or may not converge. “The morning star is the evening star” is informative at x^c because the two routes — morning observation, evening observation — are distinct intensional structures whose convergence on one object is empirical.

This is the disambiguation Kripke’s formula collapses. The formula is read at x^o ; the philosophical interest lives at x^c .

4.3 Rigidity as a Property of x^o -mode Designation

Rigidity is not an exotic semantic property names happen to have. It is what x^o designation is. A demonstrative carries no descriptive condition that could interact differently with $\Box$ depending on scope position. The tracking itself constitutes the reference. The necessity of identity falls out as a consequence of this mode of designation, not as a property of identity itself.

The *de re* / *de dicto* distinction is recovered at this level: *de re* predication is modal evaluation after the particular has been fixed (x^o mode); *de dicto* keeps the description live (x^c mode) and re-evaluates world by world. The medieval logicians had the phenomenon under a different label.

4.4 Names as the x^o/x^c Interface

BenjaminFranklin ^{o} : the man, anchored in demonstrative encounter or its traces. Rigid constitutively. *BenjaminFranklin* ^{c} : the character, anchored in descriptions, properties, stories — what one knows about him. Not rigid.

Kripke’s anti-descriptivist argument restated: names track the man (x^o), not the character (x^c). The name’s double life — rigidity from x^o , epistemic content from x^c — explains why the same name can be used by speakers with wildly different descriptive beliefs and still refer to the same individual.

“This chair” vs. “the chair”: pure x^o vs. description functioning demonstratively in context. The contrast clarifies where Kaplan’s *dthat* is needed (to rigidify the second) and where it is not (the first is already rigid).

4.5 Local Rigidity as the Primitive Form of Reference

“This chair could have been in another room” has logical form $\Diamond P(\text{this chair})$. No scope ambiguity. *This chair* does not pick out whoever satisfies some condition; it points. The reference is fixed by the perceptual encounter itself.

The inversion of the traditional order: the historical debate took descriptions as the primary model of reference and treated names as a puzzle. Local rigidity reveals the inversion. Rigidity is what reference is at its most primitive level. Descriptions are the secondary case — a workaround for situations where direct tracking is unavailable.

What baptism inherits: the scope-ambiguity-free character of x^o is what names inherit through baptism. Proper names don’t exhibit primary/secondary occurrence ambiguity because they trace back to a device that never had that ambiguity — not to a device that had it artificially suppressed.

4.6 General Term Rigidity Disambiguated

The literature’s confusion about general-term rigidity (Soames, LaPorte, Devitt) diagnosed: $water^o$ (the substance, demonstratively anchored) versus $water^c$ (the kind, descriptively anchored). Different claims, different arguments needed. The Hilary Putnam Twin-Earth case looks different once the mode is specified.

Flagged as exploratory. The full development of x^o/x^c for general terms is a research program; the dissertation argues only that the existing debate is conducted in a register where the distinction is invisible.

Chapter 5 — The PRU Account of Rigid Designation

Objective. The constructive proposal. Three-step dialectic: Russell saw the mechanism → Kripke described its semantic output → PRU explains how the mechanism scales.

5.1 The PRU Framework (Brief)

Pattern-Constellations as unified attractor states integrating sensory input and motor-affective response as one event. Pattern-Recognition Unity: recognition and reaction are inseparable aspects of a single event, not two stages. The grounding operator $\mathcal{R}(\{X\}, \langle x \rangle)$ connects the experiential dimension to the linguistic one. Only what the argument needs is introduced; the framework’s full development belongs elsewhere.

5.2 Local Rigidity: x^o as the Cognitive Primitive

x^o is the PRU implementation of Russell’s acquaintance — generalized, fallible, dynamic. $E(\{X\})$ is the experiential pole: an attractor basin shaped by prior encounters that the word x^o coordinates with in the moment of use. The disanalogy with Russell is critical. Russell required Cartesian certainty: a demonstrative referent had to be “exempt from all imaginable doubt.” PRU’s grounding is fallible — the attractor can misfire, can be resculpted by later encounters — and dynamic, not punctual.

Rigidity does not require infallibility. Kripke himself allows that we can be wrong about a bearer’s properties while still referring rigidly to it. PRU formalizes the same insight at the cognitive level: stable tracking is enough; certainty is not needed.

5.3 The Problem of Absence

Without names, rigid tracking dissolves when the bearer leaves. The demonstrative x^o is bound to an encountered instance; once the encounter ends, the binding lapses. The descriptive fallback (“the one who did such-and-such”) fails modally for the reasons Kripke gives — descriptions cannot preserve rigid tracking across counterfactual variation.

Names are *portable demonstratives*: devices engineered to export local tracking into the space of absence. They preserve what the encounter achieved, in a form that survives the end of the encounter and the entry of other speakers.

5.4 Baptism as Lifting

$\mathcal{R}(h^o, N^c)$: a baptismal act exploits an existing demonstrative binding (h^o rigidly coordinated with a particular instance of $\{H\}$) and inaugurates a new name-concept N^c whose standing function is to track that same individual.

This is not a metaphysical operation. It is a change in linguistic practice. Before baptism, the phonological form has no established role. Baptism does two things simultaneously: it exploits an existing demonstrative binding, and it inaugurates a linguistic pattern $\langle N \rangle$ whose function is to preserve that binding beyond the baptismal episode.

What shifts is the locus of rigidity, not its nature. Baptism does not *create* rigidity; it *lifts* the primitive rigidity of x^o into N^c . Names inherit the scope-ambiguity-free character of the demonstrative act at their origin.

5.5 Causal Chains as Testimonial Networks

After baptism, later speakers rarely have direct experiential contact with the bearer. Their access runs through two kinds of structure:

- **Testimonial patterns** $\tau(N)$: statements, texts, narratives produced by people who knew the bearer or learned of the bearer from those who did. The chain of teaching, citation, and correction.

- **Surrogate constellations** $\sigma(N)$: manuscripts, portraits, inscriptions, recordings — objects that causally trace back to the original bearer and offer experiential anchoring even in the bearer’s absence.

These implement Kripke’s causal chains of communication at the cognitive level. The linguistic pattern $\langle N \rangle$ is sustained by the community’s treating it as a tracking device, with norms of correction that catch deviations.

5.6 PRU as Metasemantics

PRU does not revise Kripkean model theory. In a standard Kripke model $\langle W, R, D, I \rangle$, a name N receives a rigid interpretation: $I(N)$ picks out the same individual across all worlds where that individual exists. PRU does not alter this constraint. It supplies the cognitive preconditions: how speakers come to inhabit such a model, what makes the interpretation function determinate, what the practice of rigid tracking consists in.

The relationship is parallel to the relationship between formal logic and the cognitive practices that make logical reasoning possible. The model is not in competition with the practice; the practice is what the model formalizes.

The necessary *a posteriori* is treated here as a Kripkean result the account explains. “Hesperus is Phosphorus” is necessary at x^o (one demonstrative pick, one individual) and *a posteriori* at x^c (two intensional routes whose convergence is empirical). The full machinery — $\mathcal{R}$, the two modes, baptism, testimonial chains — is in place by the time this is stated.

Chapter 6 — Objections and Replies

Objective. Address six objections, ordered roughly by depth.

6.1 “Local rigidity is just trivial deixis, not genuine modal rigidity”

The objection: x^o binding to an encountered instance is mere indexicality; it does not deliver Kripkean modal rigidity. Reply: “that bird could have been larger” is a genuine modal claim, and it requires holding *that very bird* fixed across counterfactual variation. Once $bird^o$ coordinates with a particular instance, it tracks the same individual across modal space. The mechanism is continuous with name-rigidity; the difference is scope, not kind.

6.2 “Baptism assumes what it needs to explain”

The objection: lifting h^o into N^c presupposes that there is already something to lift. Why does the lift create *persistent* rigidity, when x^o rigidity is itself temporary? Reply: persistence comes from social maintenance, not from anything intrinsic to the baptismal moment. The community treats $\langle N \rangle$ as a tracking device, with norms of correction. Wittgenstein on rule-following is the relevant background.

6.3 “This is Mental Files under a different name”

The objection: the x^o/x^c distinction is Recanati’s mental files rebranded. Reply: mental files are container metaphors — files hold information. PRU’s distinction is not about storage but about modes of word use. x^o is not a stored representation matched against perception; it is the word in its coordinating use. The container architecture is precisely what PRU rejects.

6.4 “Surrogates are irrelevant to reference transmission”

The objection: Kripke’s causal chain works through testimony alone; manuscripts and portraits add nothing semantic. Reply: surrogates provide *experiential* anchoring, not reference. Reference travels through $\langle N \rangle$. What surrogates do is keep the linguistic pattern partially grounded for speakers who otherwise have only testimony. They are not constitutive of reference but contribute to the cognitive thickness of the practice.

6.5 “The formula analysis overstates the case”

The objection: Chapter 2’s diagnosis treats Kripke’s “object” gloss as an equivocation, but Kripke is aware of the variable/instance/name distinction; the gloss is just idiomatic. Reply: the gloss does idiomatic work, but the inference from formula-validity to name-rigidity requires more than idiom. The metasemantic premise that names behave like variables is substantive and argued for separately (in *Naming and Necessity*); the formula does not establish it. The diagnostic reading shows where the substantive work lives.

6.6 “ x^o just is Russell’s acquaintance renamed”

The objection: dropping the infallibility requirement is not an innovation; it is abandoning what made acquaintance philosophically distinctive. Reply: acquaintance without infallibility is precisely what is needed to do the cognitive work the dissertation requires. Russell needed infallibility for epistemological reasons (foundationalism about knowledge); those reasons are not the dissertation’s, and the mechanism survives the change of setting. The renaming is principled, not cosmetic.

Chapter 7 — Conclusions and Future Research

Objective. Reconstruct the argument, identify the contribution, list open questions.

The Argument Reconstructed

Wittgenstein showed identity is always between designators. Kripke’s necessity-of-identity formula captures this at the x^o level, where it is trivially valid. Rus-

sell identified the mechanism (acquaintance = x^o) and retreated by demanding infallibility. Kripke described the semantic output of the mechanism for ordinary names without supplying a cognitive account. PRU supplies it: x^o as the cognitive primitive (fallible, dynamic), baptism as $\mathcal{R}(h^o, N^c)$ lifting, causal chains realized as testimonial networks with surrogate anchoring.

What PRU Explains That Others Don't

- The necessity of identity is a consequence of x^o -mode designation, not a property of identity itself.
- The necessary *a posteriori* arises because the same identity statement reads necessary at x^o and discoverable at x^c .
- The scope-ambiguity-freedom of names is inherited from the demonstrative act at their origin, not stipulated.
- The descriptions-as-rigidified case (large scope, *dthat*) is recognized as derived; rigidity at x^o is prior.

Open Questions

- Fictional names as unsaturated patterns: *Sherlock*^c with no baptismal $\mathcal{R}(h^o, N^c)$. The framework predicts that the structure of fictional reference differs from ordinary reference precisely at the lifting step.
- Natural kind terms: the x^o/x^c disambiguation as a research program. The Putnam Twin-Earth literature looks different once *water*^o and *water*^c are separated.
- Formal development of $\mathcal{R}$. The PRU framework paper sketches $\mathcal{R}$ as a projective operator with non-commutative properties; full formal treatment is parked.
- The LLM diagnostic: x^c without x^o . Whether systems that operate exclusively in $\langle x \rangle$ -space without grounding through $\mathcal{R}$ can be said to refer at all.